

Crisis Policing in Toronto’s Neighbourhoods*

Mental Health Act apprehensions are more frequent where more Indigenous and low-income residents live, and did not fall where a non-police crisis service was piloted, 2014–2025

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Toronto police detained people under Ontario’s *Mental Health Act* about 11,000 times a year between 2014 and 2025, four times in five on an officer’s own judgement. We link these 131,325 apprehensions to 2021 Census profiles of the city’s 158 neighbourhoods and to the police divisions where a non-police crisis service was piloted from 2022, and add race-based data on 65,276 arrests in 2020–2021. Apprehension rates are about 12 per cent higher for each standard-deviation increase in the Indigenous share of residents, inside and outside the downtown core, and higher where more residents have low incomes, but apprehensions did not fall in pilot divisions relative to the rest of the city. Black and Indigenous people were arrested at about three times the White rate and South Asian and East or Southeast Asian people at about half, and an officer’s mental-health flag sharply raised the chance of a strip search until a 2020 procedure change, which suggests that changing police procedure altered practice where an added crisis service has not yet.

Keywords: Mental Health Act; police apprehension; crisis response; neighbourhoods; race-based data; Toronto

1 Introduction

When someone in Toronto is in a mental health crisis, the people who arrive are usually police officers. Ontario’s *Mental Health Act* lets an officer detain a person and take them to hospital for a psychiatric assessment, a power known as an apprehension (Government of Ontario 1990). In Canada about one in three people with a mental or substance use disorder had contact with police in a year (Boyce et al. 2015), and internationally about one in ten people with a mental

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disorder had police involved in their pathway to care (Livingston 2016). Toronto police made about 10,900 apprehensions a year between 2014 and 2025 (Toronto Police Service 2026a).

Coercive routes into psychiatric care are not shared evenly. In England, compulsory admission is more likely in deprived and more ethnically diverse areas (Weich et al. 2017), and across countries Black people with a first episode of psychosis are more likely than others to reach care through police (Anderson et al. 2014). In Toronto, homeless people with mental illness have more contact with police than people who are housed (Kouyoumdjian et al. 2019), and Black residents are stopped, arrested and charged at higher rates than White residents (Wortley and Owusu-Bempah 2011; Meng 2017; Ontario Human Rights Commission 2020). Two studies use Toronto’s apprehension records directly. Gillespie et al. (2024) map apprehensions across the city’s neighbourhoods and find persistent high-rate clusters, most consistently downtown, but do not examine which neighbourhood characteristics go with high rates. The police service’s own analysis of 2021 apprehensions finds that, relative to their share of other enforcement, White people are over-represented and Black people under-represented (Toronto Police Service, n.d.). Policy has also changed. After fatal encounters, an independent review recommended changes to how police respond to people in crisis (Iacobucci 2014); a procedure change in October 2020 narrowed strip searches of people arrested, which a provincial review had found far more common in Toronto than elsewhere in Ontario (Office of the Independent Police Review Director 2019); and in 2022 the city began piloting the Toronto Community Crisis Service, which sends mental health workers instead of police to some calls (Provincial System Support Program and Shkaabe Makwa 2023). A similar program in Denver reduced reports of low-level crime (Dee and Pyne 2022), but whether Toronto’s service changed police apprehensions has not been tested.

This paper asks three questions:

1. **RQ1.** Which neighbourhood characteristics go with higher Mental Health Act apprehension rates?
2. **RQ2.** Did apprehensions fall where the Toronto Community Crisis Service was piloted?
3. **RQ3.** How do arrests and strip searches differ by perceived race, and does a mental-health flag at arrest change that?

We answer them with 131,325 apprehensions recorded by the Toronto Police Service (Toronto Police Service 2026a), linked to 2021 Census profiles of the city’s 158 neighbourhoods (City of Toronto 2026a, 2026b) and to the police divisions where the Crisis Service was piloted (Provincial System Support Program and Shkaabe Makwa 2023), and with the police service’s race-based data on 65,276 arrests in 2020 and 2021 (Toronto Police Service 2026b). For RQ1 the estimand is the association between a neighbourhood characteristic and its apprehension rate, holding the other characteristics and the year constant. For RQ2 it is the change in monthly apprehensions in pilot divisions after launch, relative to the change in other divisions over the same months. For RQ3 it is the difference in the probability of a strip search with and without an officer’s mental-instability flag, within each perceived-race group.

Apprehension rates are higher where more Indigenous people and more low-income residents live: about 12 per cent and 32 per cent higher for each standard-deviation increase in these shares. The Indigenous association holds outside the downtown core; the low-income association is weaker there, and renter share matters more. Rates are somewhat lower where more South Asian, East and Southeast Asian, or Middle-Eastern people live, and Black population share has no clear association. Apprehensions did not fall measurably in the Crisis Service’s pilot divisions: relative to other divisions, the rate ratio after launch was 1.04 (95% CI 0.81–1.33). Relative to population, Black people were arrested at 3.1 times and Indigenous people at 2.8 times the White rate, while South Asian and East and Southeast Asian people were arrested at about half the White rate. Before the 2020 procedure change, a mental-instability flag raised the probability of a strip search for most groups; after it, strip searches became rare and the gap between flagged and unflagged people almost closed.

The results show where crisis policing concentrates and which reform measurably changed police practice. They also suggest that a non-police crisis service at its current scale has not yet displaced police apprehensions, which matters as Toronto expands the service citywide (City of Toronto 2026d). A next step is to compare apprehensions with crisis calls for service, which would separate where crises occur from how police respond to them.

Section 2 describes the data and how each record is produced. Section 3 sets out the three models, Section 4 reports the results by research question, and Section 5 discusses what they mean, along with ethical considerations, weaknesses and next steps. The appendices give data details and model checks.

2 Data

2.1 Sources

Table 1 lists the four datasets; the apprehension, neighbourhood and arrest data are published on Open Data Toronto (City of Toronto 2026c) or by the police service. All were downloaded and analysed in Python (Python Software Foundation 2025) using polars (Vink et al. 2026), pandas (McKinney 2010), NumPy (Harris et al. 2020), GeoPandas (Jordahl et al. 2026), Shapely (Gillies et al. 2025) and statsmodels (Seabold and Perktold 2010), with figures made in plotnine (Kibirige et al. 2026), tables in Great Tables (Iannone and Chow 2026), tests in pytest (Krekel et al. 2026) and the paper in Quarto (Allaire et al. 2026), following the workflow of Alexander (2023).

Two other public sources were considered. Open Data Toronto publishes Persons in Crisis calls for service attended by police, coded to neighbourhoods; it measures how often police respond to a crisis rather than how often they detain someone, so we use it only to motivate next steps. The arrest data are also on Open Data Toronto, but that copy stops at 32,000 of the 65,276 rows, so we take the complete release from the police service’s own server.

Table 1: Four public datasets underpin the analysis. Records are those used after cleaning; the apprehension source holds the raw count shown in brackets, including years outside the study period. Sources: Open Data Toronto, Toronto Police Service and the Crisis Service evaluation report.

Dataset	Publisher	Unit	Period
Mental Health Act Apprehensions	Toronto Police Service	Apprehension	2014–2025
Neighbourhood Profiles and boundaries	City of Toronto	Neighbourhood	2021 Census
Crisis Service pilot areas	CAMH evaluation report	Police division	2022
Race and Identity Based Data: Arrests and Strip Searches	Toronto Police Service	Person arrested	2020–2021

2.2 Measurement: from a crisis to a record

An apprehension usually starts with a 9-1-1 call from a family member, a bystander or the person in crisis. If the officer who attends believes the person is acting in a disorderly way and is likely to seriously harm themselves or someone else, or cannot care for themselves, the officer can detain them under Section 17 of the *Mental Health Act* and take them to hospital (Government of Ontario 1990). Police also carry out apprehensions authorized by others: a physician’s application (Section 15), a justice of the peace’s order (Section 16), a community treatment order (Section 33.4), or a hospital seeking the return of an absent patient (Section 28). Each apprehension is logged in the police records system with the neighbourhood and police division where it happened, and the published data is an extract of that system.

This process shapes what the data can show. It creates **selection bias**, because only crises that reach police and end in a detention are recorded: crisis calls resolved on scene, calls diverted to the Crisis Service and voluntary visits to hospital never appear, so an apprehension measures one police response, not mental health need. It also creates **measurement bias**, because what enters the record reflects officers’ judgements. Each row is an apprehension, not a person, so someone detained five times appears five times. The location is where the apprehension happened, which need not be where the person lives. Records of people aged 17 and under are removed before publication, and neighbourhood is missing for 1,438 apprehensions and division for 854, which we drop from those analyses. The published data contains no names, exact locations or dates of birth.

The arrest data are recorded the same way, and two fields in them are officers’ judgements rather than facts about the person: race is the officer’s perception at the time of arrest, and the behaviour flags, including “mental instability or possibly suicidal”, are what the officer recorded. The arrest data give only quarters and police divisions, not dates or addresses.

2.3 Apprehensions over time and place

Apprehensions rose by about 81 per cent from 7,387 in 2014 to a peak of 13,350 in 2021, dipped after the Crisis Service began in March 2022, and have since returned to about 12,700 a year

(Figure 1). Women account for 44 per cent of apprehensions, and the most common age group is 25 to 34 (Appendix A). Of all apprehensions, 79 per cent rest on an officer’s own judgement under Section 17, up from 78 per cent in 2014 to 82 per cent in 2025.

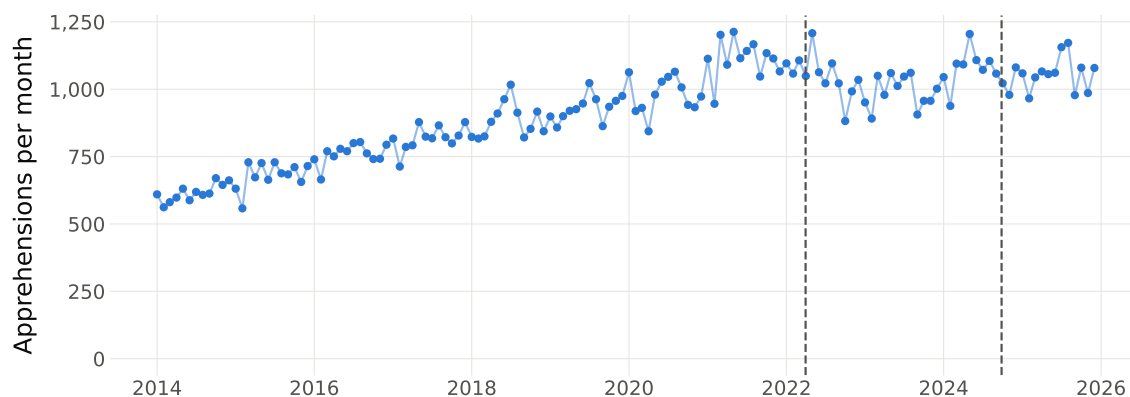


Figure 1: Mental Health Act apprehensions in Toronto rose through 2021 and have been roughly flat since. Each point is one month’s total; dashed lines mark the first Crisis Service pilot (March 2022) and its citywide launch (September 2024). Source: Toronto Police Service, Mental Health Act Apprehensions.

To compare neighbourhoods of different sizes we use rates: the average number of apprehensions per year per 1,000 residents. The median neighbourhood has 3.5, and rates run from 1.2 in Steeles to 17.1 in University. The highest rates are in the downtown core, with further pockets in the northwest and east (Figure 2). Downtown rates are inflated because apprehensions are counted where they happen but divided by the people who live there. Downtown neighbourhoods hold hospitals, shelters, transit hubs and large daytime populations, while University has only 6,435 residents. Residential population is a poor denominator for such places (Andresen 2011), so we define a downtown core of 13 neighbourhoods, those whose centre lies east of Bathurst Street, west of the Don River and south of Bloor Street (outlined in Figure 2), and repeat the main analysis without them.

2.4 Neighbourhood characteristics

Table 2 summarizes the census characteristics used in the model. Race and ethnicity shares follow the police perceived-race categories so that the two parts of the paper are comparable: East and Southeast Asian combines the Chinese, Filipino, Southeast Asian, Korean and Japanese census groups, and Middle-Eastern combines Arab and West Asian. All shares come from the 2021 Census long-form questionnaire, a 25 per cent sample, and we treat them as fixed across the 12 years. The share of households that rent ranges from under 10 per cent to almost 90 per cent, and the Indigenous share of residents, small everywhere, varies more than

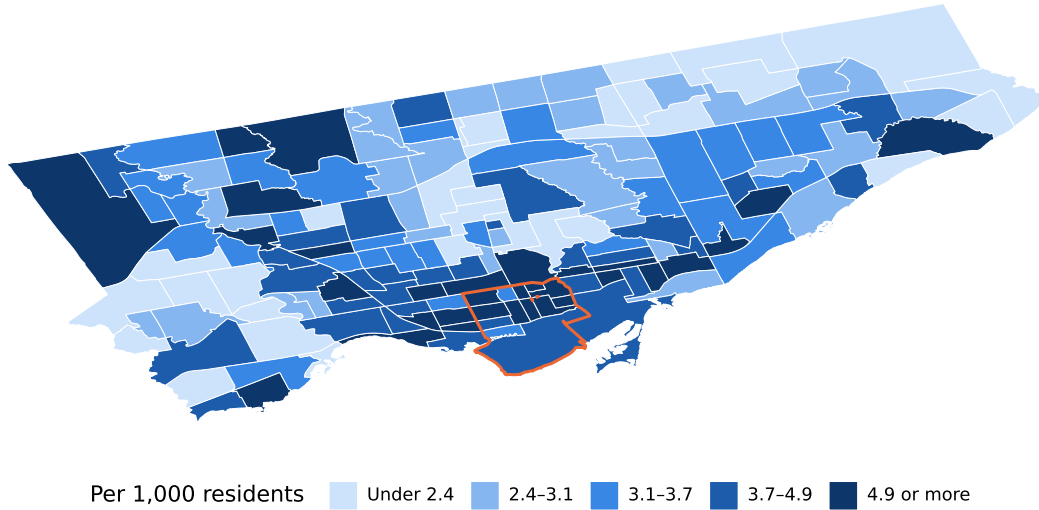


Figure 2: Apprehension rates are highest in the downtown core (outlined), with further high-rate areas in the northwest and east. Each of the 158 neighbourhoods is shaded by its average annual apprehensions per 1,000 residents in 2014–2025, grouped into fifths. Sources: Toronto Police Service; City of Toronto, Neighbourhoods and 2021 Neighbourhood Profiles.

Table 2: Toronto’s 158 neighbourhoods differ widely in renting, income and population. Values are the median and range across neighbourhoods. Sources: City of Toronto 2021 Neighbourhood Profiles; apprehension rate from Toronto Police Service, 2014–2025.

Measure	Median	Range
Apprehensions per 1,000 residents per year	3.5	1.2 – 17.1
Population	16,812	6,260 – 33,300
Median household income (\$)	84,500	57,200 – 222,000
Households that rent (%)	46	8 – 89
Residents in low income (%)	12.2	4.3 – 29.0
Indigenous residents (%)	0.8	0.0 – 3.0
Black residents (%)	6.8	0.7 – 37.9
South Asian residents (%)	7.4	2.1 – 61.6
East/Southeast Asian residents (%)	15.2	5.6 – 75.5
Middle-Eastern residents (%)	2.4	0.4 – 17.0
Latin American residents (%)	2.6	0.4 – 12.9

seventy-fold. The City designates 33 neighbourhoods as Neighbourhood Improvement Areas, its label for neighbourhoods below its equity benchmark (City of Toronto 2014).

Figure 3 plots each neighbourhood’s apprehension rate against the eight shares in the model. Rates rise with renter, low-income and Indigenous share, and with Latin American and Black share, and fall slightly with South Asian, East and Southeast Asian, and Middle-Eastern share.

Downtown neighbourhoods, in orange, sit at the top of each panel. The shares are correlated with one another (renting and low income most strongly), so the model in Section 3 estimates them together.



Figure 3: Apprehension rates rise most clearly with the share of Indigenous residents, households that rent and residents in low income. Each point is one neighbourhood; orange points are the downtown core. The vertical axis is on a log scale, and grey lines are least-squares fits on that scale. Sources: Toronto Police Service; City of Toronto 2021 Neighbourhood Profiles.

2.5 Crisis Service pilot areas

The Toronto Community Crisis Service received its first call on 31 March 2022. It was piloted in four regions defined by police division: Downtown East (divisions 51 and 52, from 31 March 2022), the Northeast (41, 42 and 43, from 4 April), Downtown West (14, from 11 July) and the Northwest (12, 23 and 31, from 18 July) (Provincial System Support Program and Shkaabe Makwa 2023). Pilot regions were chosen in part because they had the most crisis calls and apprehensions. The service went citywide on 26 September 2024 (City of Toronto 2024). The other eight divisions (11, 13, 22, 32, 33, 53, 54, 55) give a comparison group until then. Figure 4 shows that the two groups moved together before 2022: pilot divisions averaged 62 apprehensions a month from 2017 to March 2022 and 67 from April 2022 to August 2024, while other divisions went from 48 to 53.

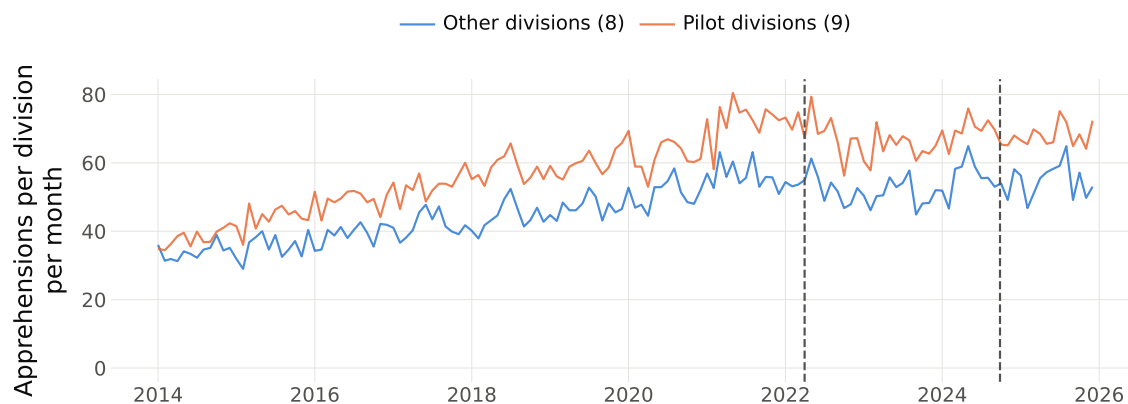


Figure 4: Apprehensions in Crisis Service pilot divisions and other divisions moved together before and after the pilots launched. Lines show the average monthly apprehensions per division in each group; dashed lines mark the first pilot launch (March 2022) and the citywide launch (September 2024). Sources: Toronto Police Service; CAMH (2023).

2.6 Arrests and strip searches by race

The arrest data contains 65,276 records for 37,347 people. To compare groups we built a population benchmark from the 2021 Census, combining census categories to match the police ones; White is approximated by residents who are neither a visible minority nor Indigenous. Counting the different people arrested at least once in 2020 and 2021 per 1,000 residents of the same perceived-race group (Figure 5), Black people were arrested at 38.6 per 1,000 and Indigenous people at 35.6, against 12.5 for White people. Middle-Eastern people (21.1) and Latino people (13.9) were also arrested at or above the White rate, while South Asian (7.3) and East and Southeast Asian people (6.0) were arrested at about half of it. The ratios are approximate, because the people arrested include non-residents while the benchmark counts residents only.

Toronto police strip searched 27 per cent of the people they arrested from January to September 2020. Strip searches are lawful only when the circumstances of an arrest give specific grounds for one (Supreme Court of Canada 2001). After a new search procedure took effect in October 2020, the rate fell to 1 per cent by the last quarter of 2021 (Appendix D). Officers flagged 3.3 per cent of arrests for “mental instability or possibly suicidal” behaviour, and this flag is the arrest data’s link to crisis policing.

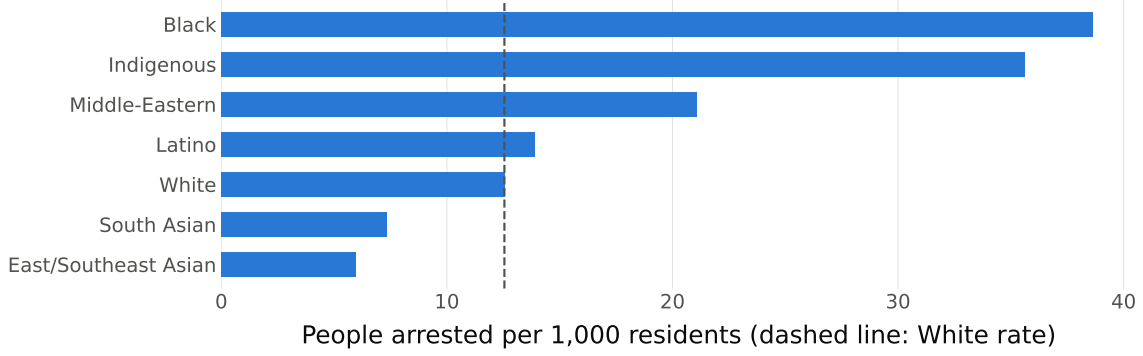


Figure 5: Relative to their share of Toronto’s population, Black and Indigenous people were arrested at about three times the White rate in 2020–2021, and South Asian and East/Southeast Asian people at about half. Bars show the number of different people arrested at least once per 1,000 residents of the same perceived-race group. Sources: Toronto Police Service race-based arrest data; 2021 Census.

3 Model

3.1 Neighbourhood apprehension rates (RQ1)

The outcome for RQ1 is the number of apprehensions y_{it} in neighbourhood i in year t . Two features of these counts shape the model. Neighbourhoods differ in size, so the model includes the population N_i as an *exposure*, which turns counts into rates. And the counts vary far more than the simplest count model, the Poisson, allows: the neighbourhood-year mean is 69 and the variance 2,582. This is called *overdispersion*, and the negative binomial distribution handles it with an extra parameter α . We therefore fit a negative binomial regression:

$$y_{it} \sim \text{NegativeBinomial}(\mu_{it}, \alpha) \quad (1)$$

$$\log \mu_{it} = \log N_i + \beta_0 + \sum_k \beta_k x_{ki} + \beta_{\text{NIA}} \text{NIA}_i + \gamma_t$$

Here μ_{it} is the expected count and α the overdispersion, so that the variance is $\mu_{it} + \alpha\mu_{it}^2$. β_0 is the log baseline rate. The x_{ki} are the eight census shares in Table 2 (renting, low income, and the Indigenous, Black, South Asian, East and Southeast Asian, Middle-Eastern and Latin American shares), each standardized to mean zero and standard deviation one across neighbourhoods so that their effects are comparable. NIA_i is 1 for a Neighbourhood Improvement Area, a designation rather than a measured quantity. The year effects γ_t absorb citywide changes, such as the pandemic and the 2021 peak, without assuming a smooth trend. Median household income is left out because it is correlated with low-income share ($r = -0.70$) and measures the same thing. Each $\exp(\beta_k)$ is a *rate ratio*: the multiplicative change in the

apprehension rate for a one-standard-deviation increase in that share, holding the others and the year fixed.

Each neighbourhood appears 12 times, and its years are not independent observations, so standard errors are *clustered* by neighbourhood: they are computed from the variation between neighbourhoods rather than treating every neighbourhood-year as separate evidence. Variance inflation factors, which measure how much the shares overlap, are all below 5. The model is fitted in statsmodels (Seabold and Perktold 2010) on 1,896 neighbourhood-years. We fit it to all neighbourhoods and, because downtown rates are inflated by non-residents, to the neighbourhoods outside the downtown core.

The model assumes that each share’s association is the same in every year and across the range of the share (on the log scale), that the 2021 census values describe neighbourhoods throughout 2014–2025, and that apprehensions are counted where residents live, which the downtown core violates. It also treats neighbourhoods as independent of their neighbours. They are not: after fitting, the residuals of neighbouring areas are correlated, with a Moran’s I (Moran 1950) of 0.16 ($p = 0.001$), where 0 would mean no correlation. Our intervals are therefore somewhat too narrow. We considered a Poisson model, which the overdispersion rules out, and a Bayesian model with a spatially structured neighbourhood effect, which would address this correlation but takes far longer to fit and check; we leave it as a next step. Appendix B reports an out-of-sample test, a comparison of fitted and observed counts, separate fits to Section 17 and other apprehensions, and a check that the model recovers known effects from simulated data.

3.2 Crisis Service pilots (RQ2)

For RQ2 the unit is a police division d in month m , from January 2017 to August 2024, before the citywide launch. We compare pilot and other divisions before and after launch, a *difference-in-differences* design:

$$\log E(y_{dm}) = a_d + c_m + \delta \text{Pilot}_d \times \text{After}_{dm} \quad (2)$$

where y_{dm} is the count of apprehensions, a_d is a division effect that absorbs each division’s usual level, c_m is a month effect that absorbs anything that changed citywide in that month, and After_{dm} is 1 from the first full month after the division’s pilot launched. $\exp(\delta)$ is the rate ratio for pilot divisions after launch, relative to what their level would have been had they followed the other divisions. Counts are negative binomial, and standard errors are clustered by division. The design assumes that, without the service, pilot and other divisions would have moved in parallel. We check this with an *event study* that replaces δ with a separate effect for each quarter before and after launch, relative to the quarter before. Pilot launches were staggered by up to four months, which can bias a single difference-in-differences estimate when effects change over time (Goodman-Bacon 2021); the event study shows whether they did.

3.3 Strip searches (RQ3)

For RQ3 we model whether booked person j was strip searched, among arrests from January to September 2020, before the procedure change, with a logistic regression:

$$\text{logit Pr}(\text{Strip}_j = 1) = \delta_0 + \delta_1 \text{Flag}_j + \sum_g \theta_g \text{Race}_{jg} + \sum_g \lambda_g (\text{Flag}_j \times \text{Race}_{jg}) + \mathbf{x}'_j \eta \quad (3)$$

The logit is the log of the odds of a strip search. Flag_j is 1 if the officer recorded mental instability. Race_{jg} is 1 if the person was perceived as belonging to group g , for each of the six groups other than White, the reference. δ_0 is the baseline log odds for an unflagged White person, δ_1 the effect of the flag for White people, θ_g the difference for group g without a flag, and λ_g how much the flag's effect differs for group g (an *interaction*). $\mathbf{x}_j$ holds offence category, age group and sex, with effects η . Standard errors are clustered by person because some people were arrested several times. We report the model as average predicted probabilities for each group with and without a flag, which are easier to read than odds ratios.

4 Results

4.1 RQ1: Apprehensions are higher where more Indigenous and low-income residents live

Table 3 and Figure 6 report the rate ratios. Across all neighbourhoods, a one-standard-deviation increase in low-income share is associated with a rate ratio of 1.32 (95% CI 1.13–1.54) and in Indigenous share 1.12 (95% CI 1.03–1.21). Renter share has a smaller association, 1.07 (95% CI 0.96–1.19). Higher Middle-Eastern (0.89 (95% CI 0.82–0.97)) and East and Southeast Asian shares (0.91 (95% CI 0.82–1.01)) go with somewhat lower rates. South Asian share, 0.97 (95% CI 0.90–1.04), Black share, 0.98 (95% CI 0.90–1.07), Latin American share, 1.01 (95% CI 0.93–1.09), and Improvement Area status, 0.92 (95% CI 0.75–1.12), are not distinguishable from no association.

Without the downtown core, the Indigenous association is unchanged, at 1.12 (95% CI 1.05–1.20). The low-income association is smaller, 1.12 (95% CI 0.99–1.27), and the renter association slightly larger, 1.10 (95% CI 1.00–1.20). The Middle-Eastern association moves towards 1, at 0.95.

Figure 7 translates three of these associations into predicted rates. Moving from a neighbourhood with an Indigenous share near the bottom of the city's range to one near the top raises the predicted rate from about 3.8 to 5.6 apprehensions per 1,000 residents a year, holding the other shares at their averages; across the range of low-income share, the predicted rate rises from about 2.9 to 8.3.

Table 3: Indigenous share is associated with higher apprehension rates with and without the downtown core; the low-income association is smaller outside it. Rate ratios per one-standard-deviation increase in each share (per 1 for Improvement Area status), with 95% confidence intervals, from negative binomial models with population exposure, year effects and standard errors clustered by neighbourhood.

	All neighbourhoods	Excluding downtown core
Households that rent	1.07 (0.96–1.19)	1.10 (1.00–1.20)
Residents in low income	1.32 (1.13–1.54)	1.12 (0.99–1.27)
Indigenous residents	1.12 (1.03–1.21)	1.12 (1.05–1.20)
Black residents	0.98 (0.90–1.07)	1.02 (0.94–1.09)
South Asian residents	0.97 (0.90–1.04)	0.96 (0.90–1.03)
East/Southeast Asian residents	0.91 (0.82–1.01)	0.93 (0.85–1.02)
Middle-Eastern residents	0.89 (0.82–0.97)	0.95 (0.88–1.01)
Latin American residents	1.01 (0.93–1.09)	1.00 (0.93–1.08)
Improvement Area	0.92 (0.75–1.12)	1.06 (0.88–1.27)
Neighbourhood-years	1,896	1,740
Overdispersion (alpha)	0.14	0.11

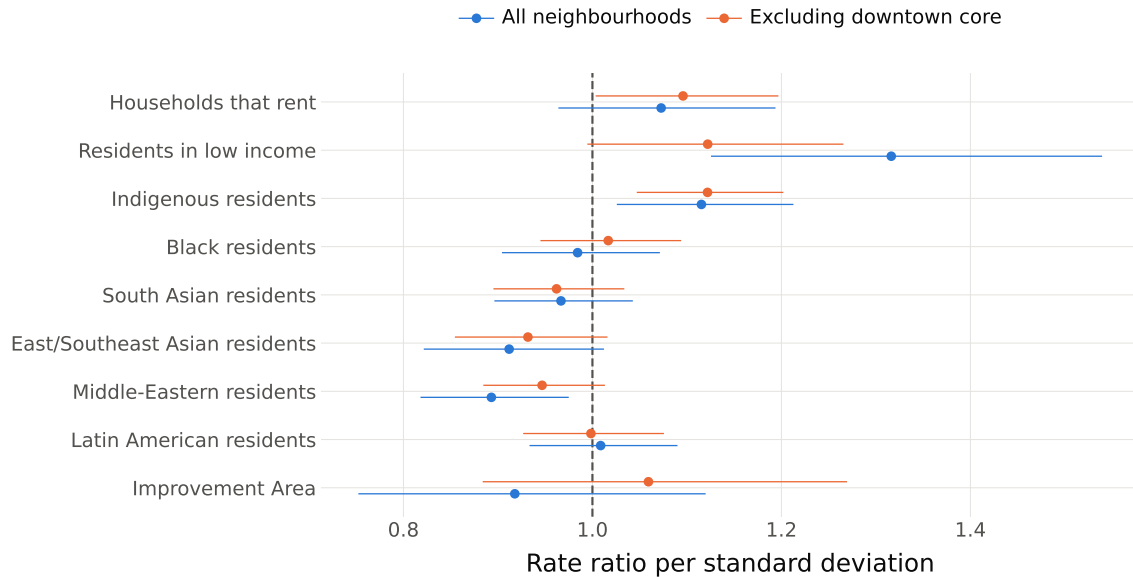


Figure 6: Indigenous and low-income shares go with higher apprehension rates, and Middle-Eastern and East/Southeast Asian shares with slightly lower ones. Points are rate ratios per one-standard-deviation increase and lines are 95% confidence intervals; values above 1 mean higher apprehension rates.

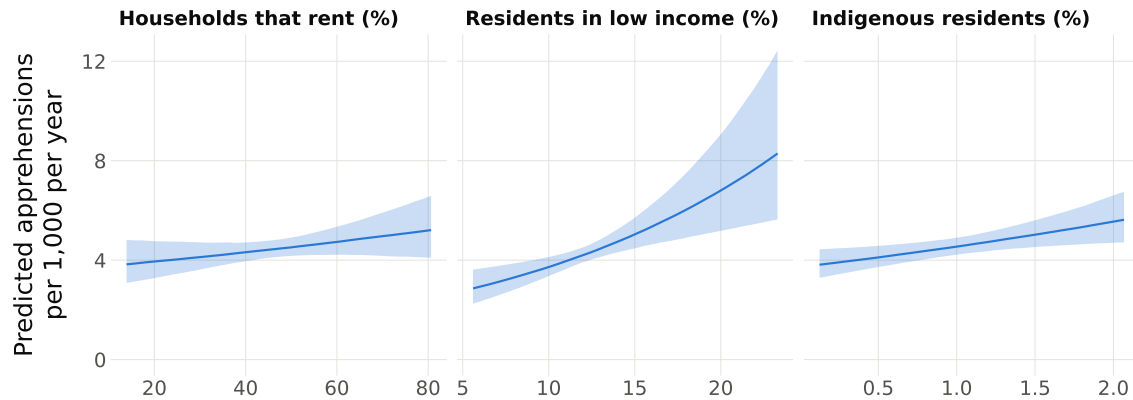


Figure 7: Predicted apprehension rates rise across the observed range of Indigenous, low-income and renter shares. Lines are predicted rates and bands 95% confidence intervals from the model for all neighbourhoods, for a neighbourhood with every other share at the city average, outside an Improvement Area, in 2025.

4.2 RQ2: Apprehensions did not fall in Crisis Service pilot divisions

Relative to other divisions, apprehensions in pilot divisions after launch had a rate ratio of 1.04 (95% CI 0.81–1.33), consistent with no change; the interval rules out reductions larger than about 19 per cent. The event study (Figure 8) supports the parallel-trends assumption: before launch, the quarterly estimates stay between 0.91 and 1.01. After launch they stay between 0.88 and 1.04, with every interval including 1.

4.3 RQ3: A mental-health flag raised strip searches until the 2020 procedure change

Among booked arrests before the procedure change, a mental-instability flag raised the odds of a strip search for White people by a factor of 2.74 (95% CI 2.11–3.56), after adjusting for offence, age and sex. In predicted probabilities (Figure 9), unflagged Black and White people were about equally likely to be strip searched, at 49 per cent and 51 per cent, and the four other groups less likely, at 37 to 38 per cent. With a flag, the probability rose to 71 per cent for White people and 79 per cent for Black people. Flagged South Asian (70 per cent) and Latino people (78 per cent) also had high predicted probabilities, but these rest on 25 and 14 flagged arrests and have wide intervals; the Indigenous estimate rests on 29. Full estimates are in Appendix C.

After the procedure change, strip searches became rare for everyone: 4.9 per cent of unflagged and 7.5 per cent of flagged booked arrests, against 47 per cent and 72 per cent before it. The gap between flagged and unflagged people fell from 25 to 3 percentage points.

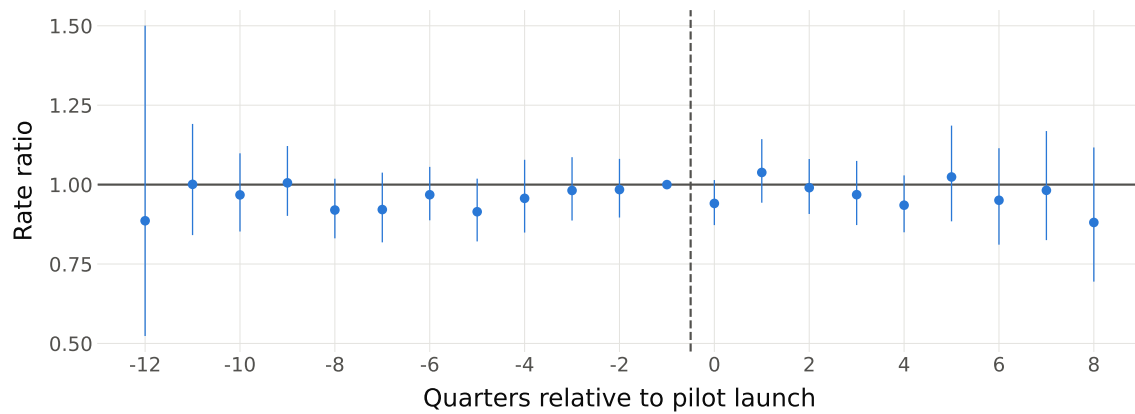


Figure 8: Apprehensions in pilot divisions tracked other divisions before and after the Crisis Service launched. Points are rate ratios for pilot divisions in each quarter relative to the quarter before launch (dashed vertical line), compared with other divisions in the same months; lines are 95% confidence intervals. The first and last points pool all earlier and later quarters. Sources: Toronto Police Service; CAMH (2023).

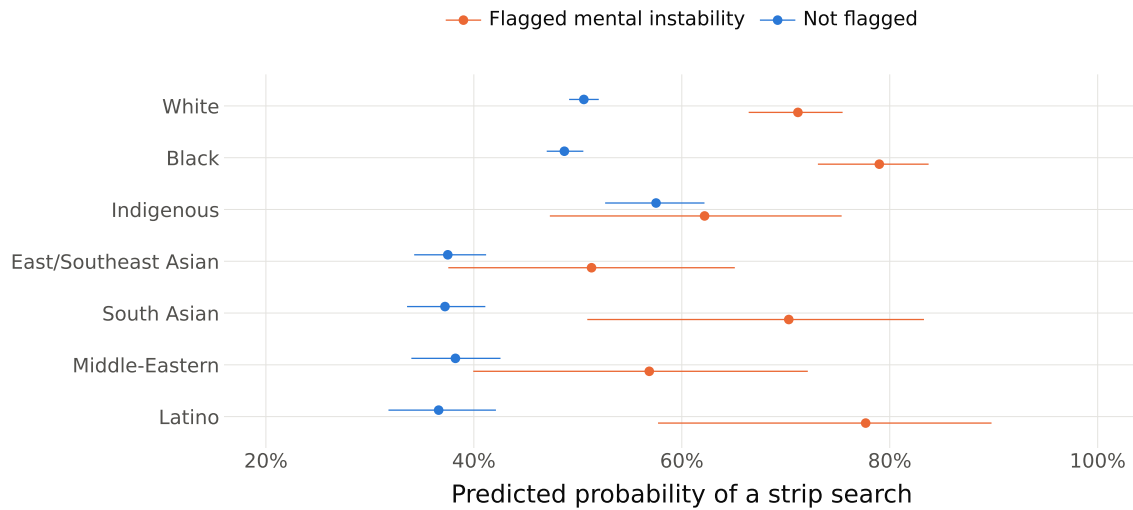


Figure 9: Before the 2020 procedure change, a mental-instability flag raised the predicted probability of a strip search for most groups. Average predicted probabilities among booked arrests in January–September 2020, adjusted for offence, age and sex; lines are 95% intervals. Groups with few flagged arrests have wide intervals. Source: Toronto Police Service race-based arrest data.

5 Discussion

5.1 What we did

We asked where crisis policing concentrates in Toronto, whether a non-police crisis service reduced it, and how crisis and race combine at arrest. We linked 12 years of police apprehension records to the census profiles of 158 neighbourhoods and modelled apprehension rates with a negative binomial regression. We compared monthly apprehensions in the police divisions where the Crisis Service was piloted with the rest of the city. And we used two years of race-based arrest data to compare arrest and strip search rates across seven perceived-race groups, with and without an officer's mental-instability flag.

5.2 Crisis policing follows Indigenous presence and poverty

The steadiest finding is that apprehensions are more frequent where more Indigenous people live. It holds inside and outside the downtown core. With neighbourhood data we cannot say whether Indigenous residents themselves are apprehended more, but the pattern matches the over-representation of Indigenous people in police contact generally, which our arrest data also show. Low income has the largest association across the whole city, but much of it comes from the downtown core, where residential rates are least reliable; outside the core, renting matters about as much. The two overlap, and renting probably stands in for things the census does not measure, such as residential instability, rooming houses and supportive housing, and fewer private resources for managing a crisis without calling 9-1-1. This fits Toronto research on homelessness and police contact (Kouyoumdjian et al. 2019) and English research on compulsory admission in deprived areas (Weich et al. 2017). The City's Improvement Area designation added nothing once these shares were in the model, so crisis policing tracks particular features of neighbourhoods more than a general measure of disadvantage.

The race and ethnicity results differ by group. Neighbourhoods with more South Asian, East and Southeast Asian, or Middle-Eastern residents have somewhat lower rates, matching the lower arrest rates of these groups in the arrest data. Black share has no clear association, although Black people are arrested at three times the White rate. This does not contradict research showing that Black people are more likely to reach psychiatric care through police (Anderson et al. 2014). The police service's own analysis suggests why: relative to their share of other enforcement, Black people are under-represented in apprehensions and White people over-represented (Toronto Police Service, n.d.). If Black people in crisis are more often arrested than apprehended, Black share would show up in arrests rather than in apprehension rates. Neighbourhood data cannot test this, but it is a reason not to read the absence of an association as the absence of disparity.

5.3 The strip search reform changed practice; the Crisis Service has not yet changed apprehensions

The two interventions we can measure had very different effects on police records. The October 2020 strip search procedure cut searches of people arrested from about a quarter to a few per cent, and the extra searching of people flagged as being in crisis largely disappeared. This is consistent with the provincial review’s finding that Toronto’s strip searches went well beyond what the law allows (Office of the Independent Police Review Director 2019; Supreme Court of Canada 2001). A change in police procedure, applied to every arrest, changed what happened at every arrest.

The Crisis Service is different. It adds an option rather than changing a rule, and police still attend most crisis calls. In 2025 it received about 4,500 eligible calls, against about 12,700 apprehensions (City of Toronto 2026d), so even if every one of its calls would otherwise have ended in an apprehension, it could have prevented only a fraction of them. Apprehensions in pilot divisions tracked the rest of the city closely before and after launch. That the service has not measurably reduced apprehensions does not mean it has failed: it may serve people who would not have been apprehended, and its own evaluation reports benefits for those it serves (Provincial System Support Program and Shkaabe Makwa 2023). But it has not yet reduced how often police detain people in crisis.

5.4 Crisis and race at arrest

The largest racial disparity in the arrest data is in who is arrested at all. Black and Indigenous people were arrested at about three times the White rate, a pattern the Ontario Human Rights Commission also found in 2013–2017, when Black people were charged at 3.9 times the White rate (Ontario Human Rights Commission 2020). South Asian and East and Southeast Asian people were arrested at about half the White rate, which shows that perceived race does not simply divide White people from everyone else. Once booked, Black and White people were strip searched at almost the same rate, and searches found items at similar rates for every group (Appendix D), which is consistent with equal search standards but does not prove them (Knowles et al. 2001; Simoiu et al. 2017). People in apparent crisis were treated differently: before the 2020 change, a mental-instability flag made a strip search far more likely in most groups. Research in Victoria, Australia, found police more likely to use or threaten weapons against people who appeared mentally ill (Kesic et al. 2013); here, apparent crisis was linked to the most intrusive search police conduct, until a procedure change removed most of the difference.

5.5 Ethics and bias

Both datasets record people at difficult moments, some of them more than once, and the analysis carries three risks. First, race in the arrest data is how police see people, which

may differ from how they identify; disparities measured this way describe police perception and practice. Second, neighbourhood results can stigmatize places. A high rate where more Indigenous people live describes where police detain people, not the people who live there, and we do not claim that Indigenous residents are apprehended more. Third, the data leave people out. A map of rates per resident shows only those counted as residents, so it hides the unhoused and visitors who make up much of downtown’s apprehensions; as D’Ignazio and Klein (2020) argue, data reflect the standpoint of whoever collects them, here a police records system. The choice of benchmark also matters: measured against population, Black people are over-represented in arrests, but measured against enforcement, they are under-represented in apprehensions (Toronto Police Service, n.d.). Each comparison answers a different question, and we have tried to state which one we answer.

5.6 Weaknesses

The neighbourhood analysis is ecological: its associations describe neighbourhoods and cannot be applied to individuals. The residential denominator distorts downtown rates, and high-traffic places outside downtown, such as the area around the Centre for Addiction and Mental Health in West Queen West, are affected too; an ambient-population denominator would be better (Andresen 2011). Census characteristics are fixed at 2021 while apprehensions span 12 years, and without a person identifier a few people detained repeatedly may weigh heavily in some neighbourhoods. Neighbouring areas’ residuals are correlated, so the neighbourhood model’s intervals are somewhat too narrow. The Crisis Service comparison has only 17 divisions, so its interval is wide; pilot areas were chosen because they had high crisis volumes, boundaries changed in late 2022, and some effects may spill into neighbouring divisions. The arrest data cover only 2020 and 2021, years shaped by the pandemic, rely on officers’ perceptions of race and behaviour, and have few flagged arrests in some groups.

5.7 Next steps

Several extensions would address these weaknesses. A model with a spatially structured neighbourhood effect would account for similarity between neighbouring areas (Riebler et al. 2016). Persons in Crisis calls for service, which Open Data Toronto publishes by neighbourhood, would allow apprehensions to be measured per crisis call, separating where crises occur from how police respond. Linking apprehensions to health records, as Ontario studies of involuntary admission have done (Lebenbaum et al. 2018), would show who is detained and what happens next. Once the citywide Crisis Service has run for longer, comparing neighbourhoods by how many calls it diverts would test its effect more directly than the division comparison here. And publishing apprehension data with the person’s race, which the police service already collects (Toronto Police Service, n.d.), would allow the individual-level analysis that neighbourhood data cannot support.

Declarations

Data and code availability. All data are public: the apprehension, neighbourhood and census data from Open Data Toronto under the Open Government Licence – Toronto, and the arrest data from the Toronto Police Service. Code, cleaned data, a snapshot of the raw data and the fitted models are at <https://github.com/TheZarif/toronto-crisis-policing> under the MIT License.

Use of generative AI. Claude (Anthropic) assisted with analysis code, documentation, the literature search and revision of the text. The author checked every citation against its source and every result against the code, and takes responsibility for the content.

References

- Alexander, Rohan. 2023. *Telling Stories with Data*. Chapman and Hall/CRC. <https://tellingstorieswithdata.com/>.
- Allaire, J. J., Charles Teague, Carlos Scheidegger, Yihui Xie, Christophe Dervieux, and Gordon Woodhull. 2026. *Quarto*. <https://doi.org/10.5281/zenodo.5960048>.
- Anderson, Kelly K., Nina Flora, Suzanne Archie, Craig Morgan, and Kwame McKenzie. 2014. “A Meta-Analysis of Ethnic Differences in Pathways to Care at the First Episode of Psychosis.” *Acta Psychiatrica Scandinavica* 130: 257–68. <https://doi.org/10.1111/acps.12254>.
- Andresen, Martin A. 2011. “The Ambient Population and Crime Analysis.” *The Professional Geographer* 63 (2): 193–212. <https://doi.org/10.1080/00330124.2010.547151>.
- Boyce, Jillian, Cristine Rotenberg, and Maisie Karam. 2015. “Mental Health and Contact with Police in Canada, 2012.” *Juristat*. <https://www150.statcan.gc.ca/n1/pub/85-002-x/2015001/article/14176-eng.htm>.
- City of Toronto. 2014. *TSNS 2020 Neighbourhood Equity Index: Methodological Documentation*. City of Toronto. <https://www.toronto.ca/wp-content/uploads/2017/11/97eb-TSNS-2020-NEI-equity-index-methodology-research-report-backgroundfile-67350.pdf>.
- City of Toronto. 2024. *Mayor Olivia Chow Launches Toronto Community Crisis Service City-Wide*. <https://www.toronto.ca/news/mayor-olivia-chow-launches-toronto-community-crisis-service-city-wide/>.

- City of Toronto. 2026a. *Neighbourhood Profiles*. <https://open.toronto.ca/dataset/neighbourhood-profiles/>.
- City of Toronto. 2026b. *Neighbourhoods*. <https://open.toronto.ca/dataset/neighbourhoods/>.
- City of Toronto. 2026c. *Open Data Toronto*. <https://open.toronto.ca/>.
- City of Toronto. 2026d. *Toronto Community Crisis Service: Growing Toronto’s Community-Based Crisis Response System*. Report for action to Executive Committee. City of Toronto. <https://www.toronto.ca/legdocs/mmis/2026/ex/bgrd/backgroundfile-289503.pdf>.
- D’Ignazio, Catherine, and Lauren F. Klein. 2020. *Data Feminism*. MIT Press. <https://data-feminism.mitpress.mit.edu/>.
- Dee, Thomas S., and Jaymes Pyne. 2022. “A Community Response Approach to Mental Health and Substance Abuse Crises Reduced Crime.” *Science Advances* 8 (23): eabm2106. <https://doi.org/10.1126/sciadv.abm2106>.
- Gillespie, Alexandra, Jennifer Stringer, and Olaf Berke. 2024. “Spatial Patterns of Mental Health Act Apprehensions in Toronto, Canada, 2014–2022.” *Canadian Journal of Public Health* 115 (6): 862–79. <https://doi.org/10.17269/s41997-024-00912-7>.
- Gillies, Sean et al. 2025. *Shapely: Manipulation and Analysis of Geometric Objects*. <https://shapely.readthedocs.io/>.
- Goodman-Bacon, Andrew. 2021. “Difference-in-Differences with Variation in Treatment Timing.” *Journal of Econometrics* 225 (2): 254–77. <https://doi.org/10.1016/j.jeconom.2021.03.014>.
- Government of Ontario. 1990. *Mental Health Act, R.S.O. 1990, c. M.7*. <https://www.ontario.ca/laws/statute/90m07>.
- Harris, Charles R., K. Jarrod Millman, Stéfan J. van der Walt, et al. 2020. “Array Programming with NumPy.” *Nature* 585: 357–62. <https://doi.org/10.1038/s41586-020-2649-2>.
- Iacobucci, Frank. 2014. *Police Encounters with People in Crisis: An Independent Review Conducted by the Honourable Frank Iacobucci for Chief of Police William Blair, Toronto Police Service*. Independent review. Toronto Police Service.
- Iannone, Richard, and Michael Chow. 2026. *Great Tables: Make Awesome Display Tables Using Python*. <https://posit-dev.github.io/great-tables/>.
- Jordahl, Kelsey et al. 2026. *GeoPandas: Python Tools for Geographic Data*. <https://geopandas>.

org/.

- Kesic, Dragana, Stuart D. M. Thomas, and James R. P. Ogloff. 2013. "Use of Nonfatal Force on and by Persons with Apparent Mental Disorder in Encounters with Police." *Criminal Justice and Behavior* 40 (3): 321–37. <https://doi.org/10.1177/0093854812474425>.
- Kibirige, Hassan et al. 2026. *Plotnine: A Grammar of Graphics for Python*. <https://plotnine.org/>.
- Knowles, John, Nicola Persico, and Petra Todd. 2001. "Racial Bias in Motor Vehicle Searches: Theory and Evidence." *Journal of Political Economy* 109 (1): 203–29. <https://doi.org/10.1086/318603>.
- Kouyoumdjian, Fiona G., Ri Wang, Cilia Mejia-Lancheros, et al. 2019. "Interactions Between Police and Persons Who Experience Homelessness and Mental Illness in Toronto, Canada: Findings from a Prospective Study." *The Canadian Journal of Psychiatry* 64 (10): 718–25. <https://doi.org/10.1177/0706743719861386>.
- Krekel, Holger et al. 2026. *Pytest*. <https://pytest.org/>.
- Lebenbaum, Michael, Maria Chiu, Simone Vigod, and Paul Kurdyak. 2018. "Prevalence and Predictors of Involuntary Psychiatric Hospital Admissions in Ontario, Canada: A Population-Based Linked Administrative Database Study." *BJPsych Open* 4 (2): 31–38. <https://doi.org/10.1192/bjo.2017.4>.
- Livingston, James D. 2016. "Contact Between Police and People with Mental Disorders: A Review of Rates." *Psychiatric Services* 67 (8): 850–57. <https://doi.org/10.1176/appi.ps.201500312>.
- McKinney, Wes. 2010. "Data Structures for Statistical Computing in Python." *Proceedings of the 9th Python in Science Conference*, 56–61. <https://doi.org/10.25080/Majora-92bf1922-00a>.
- Meng, Yunliang. 2017. "Profiling Minorities: Police Stop and Search Practices in Toronto, Canada." *Human Geographies* 11 (1): 5–23. <https://doi.org/10.5719/hgeo.2017.111.1>.
- Morabito, Melissa Schaefer. 2007. "Horizons of Context: Understanding the Police Decision to Arrest People with Mental Illness." *Psychiatric Services* 58 (12): 1582–87. <https://doi.org/10.1176/ps.2007.58.12.1582>.
- Moran, P. A. P. 1950. "Notes on Continuous Stochastic Phenomena." *Biometrika* 37 (1/2): 17–23. <https://doi.org/10.2307/2332142>.

- Office of the Independent Police Review Director. 2019. *Breaking the Golden Rule: A Review of Police Strip Searches in Ontario*. Office of the Independent Police Review Director.
- Ontario Human Rights Commission. 2020. *A Disparate Impact: Second Interim Report on the Inquiry into Racial Profiling and Racial Discrimination of Black Persons by the Toronto Police Service*. Ontario Human Rights Commission. <https://www.ohrc.on.ca/en/disperate-impact-second-interim-report-inquiry-racial-profiling-and-racial-discrimination-black>.
- Provincial System Support Program, and Shkaabe Makwa. 2023. *Toronto Community Crisis Service: One-Year Outcome Evaluation Report*. Centre for Addiction; Mental Health. <https://www.toronto.ca/wp-content/uploads/2023/10/97b8-Toronto-Community-Crisis-Service-1-Year-Outcome-Evaluation-Report.pdf>.
- Python Software Foundation. 2025. *Python: A Programming Language*. <https://www.python.org/>.
- Riebler, Andrea, Sigrunn H. Sørbye, Daniel Simpson, and Håvard Rue. 2016. “An Intuitive Bayesian Spatial Model for Disease Mapping That Accounts for Scaling.” *Statistical Methods in Medical Research* 25 (4): 1145–65. <https://doi.org/10.1177/0962280216660421>.
- Seabold, Skipper, and Josef Perktold. 2010. “Statsmodels: Econometric and Statistical Modeling with Python.” *Proceedings of the 9th Python in Science Conference*, 92–96. <https://doi.org/10.25080/Majora-92bf1922-011>.
- Simoiu, Camelia, Sam Corbett-Davies, and Sharad Goel. 2017. “The Problem of Infra-Marginality in Outcome Tests for Discrimination.” *The Annals of Applied Statistics* 11 (3): 1193–216. <https://doi.org/10.1214/17-AOAS1058>.
- Supreme Court of Canada. 2001. *R v Golden, 2001 SCC 83, [2001] 3 SCR 679*. <https://www.canlii.org/en/ca/scc/doc/2001/2001scc83/2001scc83.html>.
- Toronto Police Service. 2022. *Race & Identity Based Data Collection Strategy: Understanding Use of Force & Strip Searches in 2020, Detailed Report*. Toronto Police Service. <https://www.tps.ca/race-identity-based-data-collection/>.
- Toronto Police Service. 2026a. *Mental Health Act Apprehensions*. <https://open.toronto.ca/dataset/mental-health-apprehensions/>.
- Toronto Police Service. 2026b. *Race & Identity Based Data Collection: Arrests and Strip Searches*. https://services.arcgis.com/S9th0jAJ7bqgIRjw/ArcGIS/rest/services/RBDC_ARR_TBL_001/FeatureServer.
- Toronto Police Service. n.d. *Mental Health Apprehensions: Sharing Preliminary Find-*

ings. <https://www.tps.ca/race-identity-based-data-collection/follow-our-progress/mental-health-apprehensions-sharing-preliminary-findings/>.

Vink, Ritchie et al. 2026. *Polars: Blazingly Fast DataFrames in Rust, Python, Node.js, R and SQL*. <https://pola.rs/>.

Weich, Scott, Orla McBride, Liz Twigg, et al. 2017. “Variation in Compulsory Psychiatric Inpatient Admission in England: A Cross-Classified, Multilevel Analysis.” *The Lancet Psychiatry* 4 (8): 619–26. [https://doi.org/10.1016/S2215-0366\(17\)30207-9](https://doi.org/10.1016/S2215-0366(17)30207-9).

Wortley, Scot, and Akwasi Owusu-Bempah. 2011. “The Usual Suspects: Police Stop and Search Practices in Canada.” *Policing and Society* 21: 395–407. <https://doi.org/10.1080/10439463.2011.610198>.

Appendix

Appendix A Data details

A.1 Cleaning

- **Period.** We keep apprehensions that occurred from 2014 to the last complete calendar year, 2025, dropping 13 records with earlier occurrence dates and the partial current year.
- **Missing codes.** “NSA” and “Not Recorded” values are treated as missing.
- **Repeated incidents.** Some incidents involve more than one person and share an incident number. All rows are kept because each is a separate apprehension, and there is no person identifier with which to identify duplicates.
- **Panels.** Every neighbourhood-year and division-month appears, with zeros where there were no apprehensions.
- **Downtown core.** Neighbourhoods whose centroid lies east of Bathurst Street (longitude -79.4065), west of the Don River (-79.356) and south of Bloor Street (latitude 43.6705), approximating the City’s Downtown Plan area.
- **Arrest data.** Labels that changed between 2020 and 2021 (offence categories, age groups, youth status) are combined into one scheme. Strip searches without a recorded booking are treated as booked, following the source documentation. Our cleaned data reproduces the Toronto Police Service’s published strip search rate of 22.2 per cent for 2020 (Toronto Police Service 2022).
- **Tests.** Automated tests check every cleaned and simulated table for schema, valid values, internal consistency and agreement with the raw data.

A.2 The data

Table 4 shows five rows of the cleaned apprehension data, Table 5 counts apprehensions by legal authority, and Figure 10 shows who is apprehended. Table 6 summarizes the arrest variables used in the strip search model.

Table 4: Each row of the cleaned apprehension data is one apprehension. Five randomly chosen rows, with event identifiers removed. Source: Toronto Police Service.

Date	Division	Neighbourhood	Authority	Place	Sex	Age group
2016-03-27	D43	Bendale South	Section 17	House	Male	25 to 34
2018-08-01	D33	Hillcrest Village	Section 16	House	Male	25 to 34
2021-02-05	D42	Tam O’Shanter-Sullivan	Section 17	Apartment	Male	35 to 44
2021-03-02	D14	West Queen West	Section 15	Other	Female	55 to 64
2024-05-18	D52	Bay-Cloverhill	Section 15	Other	Female	25 to 34

Table 5: Four in five apprehensions in 2014–2025 were made under Section 17, on the officer’s own judgement. The other authorities carry out a decision by a physician, a justice of the peace or a hospital. Source: Toronto Police Service.

Authority	Who decides	Apprehensions	Share
Section 17	Police officer, on scene	104,164	79.3%
Section 15 (Form 1)	Physician	10,703	8.2%
Section 16 (Form 2)	Justice of the peace	9,074	6.9%
Section 33.4 (Form 47)	Physician, community treatment order	5,777	4.4%
Section 28 (Form 9)	Hospital, patient absent without leave	1,607	1.2%

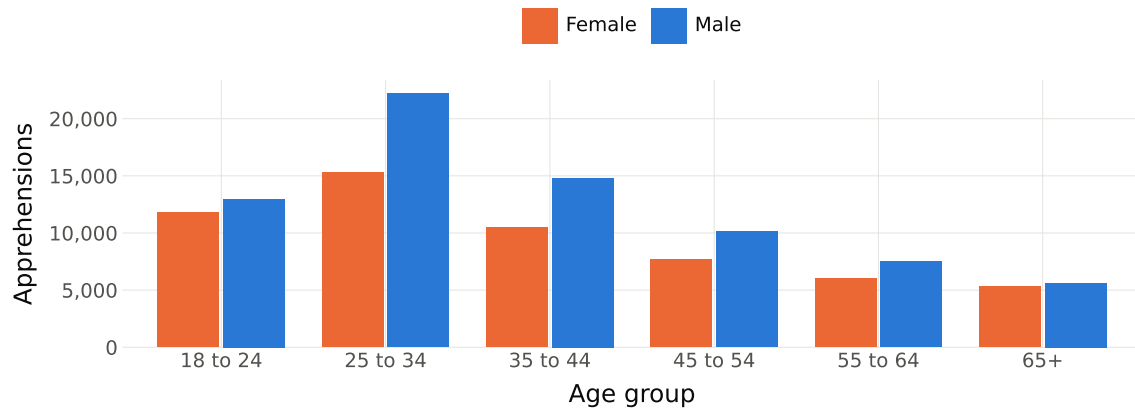


Figure 10: Men outnumber women among people apprehended in every age group, and people aged 25 to 34 are apprehended most often. Bars count all apprehensions in 2014–2025 with sex and age recorded. Source: Toronto Police Service.

Table 6: The strip search model uses booked arrests from January to September 2020. Shares are within each perceived-race group. The most common offence category is warrants, compliance and administrative charges for White, Black, Indigenous and Middle-Eastern people, and assault for South Asian, East/Southeast Asian and Latino people. Source: Toronto Police Service race-based arrest data.

Perceived race	Booked arrests	Male	Aged 18–34	Assault	Flagged	Strip searched
White	5,930	80%	43%	23%	5%	52%
Black	3,983	86%	61%	24%	6%	53%
Indigenous	455	74%	53%	26%	6%	59%
East/Southeast Asian	807	88%	50%	34%	5%	35%
South Asian	702	89%	50%	31%	4%	33%
Middle-Eastern	565	91%	54%	27%	6%	35%
Latino	357	87%	52%	34%	4%	33%

Appendix B Model details

B.1 Model checks

To test the model out of sample, we refit it to 2014–2023 and predicted 2024 and 2025 using the 2023 year effect. Its root mean squared error was 34.0 apprehensions per neighbourhood-year and its mean absolute error 22.8, against an average of 80 apprehensions per neighbourhood-year in those years. Figure 11 compares fitted with observed counts for every neighbourhood-year. Points lie close to the diagonal across the range, with the largest residuals in a few downtown neighbourhoods.



Figure 11: The model’s fitted counts track the observed counts across all 1,896 neighbourhood-years. Each point is one neighbourhood in one year; orange points are the downtown core; the dashed line marks perfect agreement. Both axes are on a log scale.

B.2 Apprehension types

Table 7 repeats the model for all neighbourhoods and outside the downtown core, and adds separate fits to apprehensions made under Section 17, on an officer’s own judgement, and those authorized by a physician, a justice of the peace or a hospital. If officers’ discretion were shaped by the character of places (Morabito 2007), Section 17 apprehensions would track neighbourhood characteristics more closely; they do not.

Table 7: The associations are similar for apprehensions made on an officer’s own judgement and those authorized by others. Rate ratios per one-standard-deviation increase (per 1 for Improvement Area status) with 95% confidence intervals, from negative binomial models with population exposure, year effects and standard errors clustered by neighbourhood.

	All	Excl. downtown	Section 17	Other types
Households that rent	1.07 (0.96–1.19)	1.10 (1.00–1.20)	1.08 (0.99–1.18)	1.03 (0.82–1.29)
Residents in low income	1.32 (1.13–1.54)	1.12 (0.99–1.27)	1.27 (1.10–1.46)	1.50 (1.15–1.95)
Indigenous residents	1.12 (1.03–1.21)	1.12 (1.05–1.20)	1.10 (1.03–1.18)	1.19 (0.99–1.43)
Black residents	0.98 (0.90–1.07)	1.02 (0.94–1.09)	0.99 (0.92–1.06)	1.00 (0.85–1.17)
South Asian residents	0.97 (0.90–1.04)	0.96 (0.90–1.03)	0.98 (0.91–1.04)	0.93 (0.81–1.06)
East/Southeast Asian residents	0.91 (0.82–1.01)	0.93 (0.85–1.02)	0.92 (0.83–1.01)	0.90 (0.76–1.07)
Middle-Eastern residents	0.89 (0.82–0.97)	0.95 (0.88–1.01)	0.90 (0.84–0.98)	0.86 (0.74–1.00)
Latin American residents	1.01 (0.93–1.09)	1.00 (0.93–1.08)	1.02 (0.95–1.08)	0.98 (0.83–1.16)
Improvement Area	0.92 (0.75–1.12)	1.06 (0.88–1.27)	0.94 (0.78–1.13)	0.84 (0.60–1.17)

B.3 Recovery of simulated effects

Before fitting the models to real data, we fitted them to simulated data in which the true effects are known. In the simulated neighbourhoods, the log apprehension rate rises by 0.015 per percentage point of renter households and falls by 0.6 per unit of log median income, with overdispersion $\alpha = 0.2$; in the simulated divisions, apprehensions fall by 15 per cent in pilot divisions after launch. The models recover these values (Table 8).

Table 8: The count and difference-in-differences specifications recover the effects built into the simulated data. True values are those used to simulate; estimates and 95% confidence intervals are from fitting the models to the simulation.

Parameter	True value	Estimate	95% CI
Renter share (per point)	0.015	0.015	0.013 to 0.016
Log median income	-0.60	-0.61	-0.70 to -0.52
Overdispersion (alpha)	0.20	0.19	
Pilot effect (rate ratio)	0.85	0.87	0.81 to 0.92

Appendix C Strip search model

Table 9: A mental-instability flag more than doubles the odds of a strip search for White people; the interaction terms show how the flag’s effect differs for other groups, with wide intervals where few arrests were flagged. Odds ratios with 95% confidence intervals from a logistic regression of strip searching among booked arrests in January–September 2020, adjusted for offence category, age group and sex, with standard errors clustered by person. White people are the reference group.

Term	Odds ratio	95% CI low	95% CI high	p-value
Black	0.92	0.82	1.02	0.111
East/Southeast Asian	0.53	0.44	0.64	<0.001
Indigenous	1.38	1.09	1.76	0.008
Latino	0.51	0.39	0.67	<0.001
Middle-Eastern	0.55	0.44	0.70	<0.001
South Asian	0.53	0.43	0.65	<0.001
Mental-instability flag	2.74	2.11	3.56	<0.001
Flag × Black	1.75	1.11	2.74	0.015
Flag × East/Southeast Asian	0.71	0.35	1.43	0.334
Flag × Indigenous	0.46	0.20	1.02	0.056
Flag × Latino	2.88	0.94	8.80	0.064
Flag × Middle-Eastern	0.88	0.37	2.09	0.778
Flag × South Asian	1.81	0.72	4.56	0.209

Appendix D Disparities at each stage of contact

Figure 12 shows strip search rates among booked arrests before and after the October 2020 procedure change. Table 10 follows each perceived-race group from the population through arrest, booking and strip search before the change, and reports how often strip searches found items. The largest departure from population shares is at arrest. Booking and strip search rates for Black and White people are similar, and the share of strip searches that found items is similar for every group.

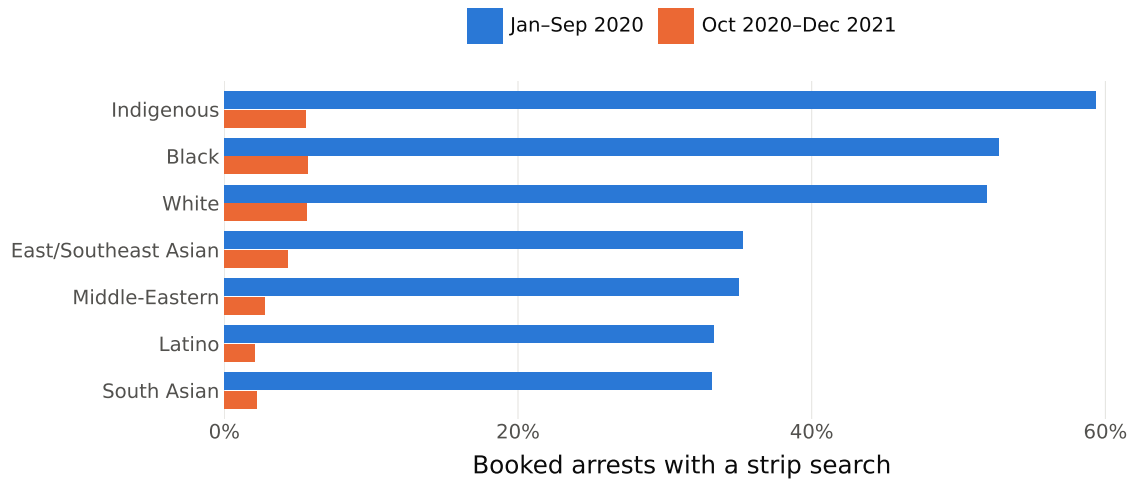


Figure 12: Before the October 2020 procedure change, more than half of booked Indigenous, Black and White people were strip searched; afterwards rates fell below 6 per cent for every group. Bars show the share of booked arrests that included a strip search. Source: Toronto Police Service race-based arrest data.

Table 10: The largest racial disparity is at arrest; booking, strip search and search outcomes differ much less between Black and White people. Population share is from the 2021 Census; arrest share counts different people arrested in 2020–2021; booking and strip search rates are for January–September 2020; items found is the share of all strip searches in 2020–2021 that found items. Sources: Toronto Police Service race-based arrest data; 2021 Census.

Perceived race	Population	People arrested	Booked	Strip searched	Items found
White	45%	42%	56%	52%	38%
East/Southeast Asian	22%	10%	52%	35%	34%
South Asian	15%	8%	54%	33%	35%
Black	10%	28%	59%	53%	38%
Middle-Eastern	4%	7%	50%	35%	37%
Latino	3%	4%	60%	33%	36%
Indigenous	1%	2%	61%	59%	38%